Reproducing and Evaluating HVI-CIDNet: A New Color Space for Low-Light Image Enhancement

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Abstract—Low-light image enhancement (LLIE) is a fundamental low-level vision task that aims to restore visibility, contrast, and color fidelity in images captured under poor illumination, while avoiding noise amplification and color distortion. This report presents a reproduction and independent evaluation of HVI-CIDNet (Yan et al., CVPR 2025), a recent LLIE method that introduces the Horizontal/Vertical-Intensity (HVI) color space together with a dual-branch Color and Intensity Decoupling Network (CIDNet) guided by Lighten Cross-Attention (LCA) blocks. We rebuilt the official open-source implementation from scratch in a clean, CPU-capable, reproducible environment, resolved several non-trivial dependency and platform-compatibility issues that the original repository does not document, and packaged the result as a self-contained Gradio demo. Using the authors' officially released generalization checkpoint, we reproduce the qualitative low-light enhancement behavior of HVI-CIDNet and conduct a supplementary no-reference quality evaluation (NIQE, BRISQUE) on a synthetically darkened test set derived from public-domain images, since the original paired/unpaired benchmark datasets are only distributed through region-locked file-sharing links. Our reproduction lowers the average NIQE from 10.86 to 4.43 and the average BRISQUE from 17.98 to 6.89 across eight test images, consistent in direction with the officially reported improvements on the DICM/LIME/MEF/NPE/VV benchmarks. We discuss the engineering challenges of reproducing a 2023-era PyTorch/Python 3.7 research codebase on a modern Windows machine, and outline limitations and directions for a deeper, dataset-faithful evaluation.

Index Terms—low-light image enhancement, color space, attention, image restoration, reproducibility, HVI-CIDNet

I. Introduction

Images captured under low-light conditions – at night, indoors with poor lighting, or with short exposure times – typically suffer from low contrast, color shift, and severe sensor noise. Low-light image enhancement (LLIE) seeks to recover an image that resembles what would have been captured under normal illumination, which is valuable both as a standalone photographic tool and as a pre-processing step for downstream vision tasks such as detection and segmentation that otherwise degrade sharply in the dark.

Classical approaches to LLIE include histogram equalization and its variants, and physically motivated Retinex-based decompositions that split an image into illumination and reflectance components [1]. With the rise of deep learning, learned LLIE models have substantially outperformed classical methods by directly learning a mapping from low-light to normal-light images using paired datasets such as LOL [1] or by exploiting zero-reference and unpaired supervision when paired data is unavailable [2], [3]. More recently, transformer-based restoration backbones [4] and Retinex-guided transformers [5] have pushed the state of the art further, at the cost of increased model and training complexity.

A common thread across most of these methods is that they operate directly in the RGB color space, which couples brightness and color information in a way that makes it difficult to enhance illumination without introducing color artifacts or amplifying noise in saturated channels. HVI-CIDNet [6], [7] addresses this by proposing a new Horizontal/Vertical-Intensity (HVI) color space that compresses the saturation/hue components into a bounded, polar representation while keeping a separate, photometrically meaningful intensity channel. This decoupling lets a dual-branch network – the Color and Intensity Decoupling Network (CIDNet) – process color and brightness with separate encoder-decoder pathways that exchange information through learned Lighten Cross-Attention (LCA) blocks, rather than forcing a single RGB branch to disentangle both implicitly. HVI-CIDNet was accepted to CVPR 2025 and reports state-of-the-art or near state-of-the-art PSNR/SSIM/LPIPS on the LOLv1, LOLv2-real, LOLv2-synthetic, Sony Total Dark (SID), LOL-Blur, SICE, and FiveK benchmarks, as well as strong no-reference quality scores (NIQE, BRISQUE) on five widely used unpaired datasets (DICM, LIME, MEF, NPE, VV).

The goal of this course project is not to propose a new LLIE method, but to critically reproduce HVI-CIDNet end-to-end: clone the official repository, build a working Python environment, obtain the officially released model weights, verify that the qualitative and quantitative behavior reported by the authors can actually be reproduced by an independent party, and package the result as a usable, documented demo. This kind of reproduction work is explicitly valuable to the community: published repositories frequently rot due to unpinned dependencies, and a method's real-world usability depends as much on whether

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a newcomer can actually get it running as on its reported benchmark numbers. Concretely, our contributions are:

- A documented, from-scratch rebuild of the HVI-CIDNet environment on a machine with no pre-existing Python/conda installation, including the resolution of several dependency-compatibility issues (Section IV-D) that are not mentioned in the official repository's documentation;
- A packaged, CPU-capable Gradio demo with the officially recommended generalization.pth checkpoint (obtained via the Hugging Face Hub) pre-loaded, with usage instructions;
- An independent supplementary evaluation using no-reference image quality metrics (NIQE, BRISQUE) on a synthetically generated low-light test set, cross-checked against the authors' officially published numbers on the five standard unpaired benchmarks;
- A discussion of the gap between "the method works in the paper" and "the method runs on my machine," framed as a set of concrete, reproducible engineering lessons.

The remainder of this report is organized as follows. Section II formalizes the LLIE problem and reviews the evaluation metrics and standard datasets used throughout this report. Section III surveys related work in low-light image enhancement. Section IV describes the HVI color space, the CIDNet architecture, and our reproduction methodology, including the practical issues we encountered and resolved. Section V presents our experimental analysis, combining the authors' officially reported results with our own supplementary evaluation. Section VI concludes and discusses limitations and future work.

II. Background: Problem Formulation and Evaluation Metrics

This section formalizes the LLIE task and reviews the evaluation metrics and benchmark datasets referenced throughout the rest of this report, so that the quantitative claims in Section V can be read without consulting external sources.

A. Problem Formulation

A simplified image formation model treats a captured low-light image $x \in [0,1]^{3\times H\times W}$ as a degraded version of a hypothetical normal-light image y of the same scene,

$$x = f(y) \odot t + n, \qquad t \ll 1, \tag{1}$$

where t is a (possibly spatially varying) transmission/exposure factor that is small under low illumination, $f(\cdot)$ is a non-linear camera response function, $\odot$ denotes element-wise multiplication, and n is signal-dependent sensor noise that becomes visually dominant precisely because t is small (the same absolute noise power is much more visible once the signal itself has been attenuated). LLIE seeks a function g_θ, implemented here as the CIDNet network of Section IV, such that $g_\theta(x) \approx y$. Because t, f, and n are unknown and scene-dependent at test time, LLIE is an ill-posed inverse problem: a single dark observation x is consistent with many plausible normal-light reconstructions y (the same ambiguity that motivates probabilistic formulations such as LLFlow [8]). Retinex-based methods make this tractable by further factorizing $y = r \cdot \ell$ into a reflectance map r (assumed illumination-invariant) and an illumination map ℓ, and only attempting to correct ℓ rather than regressing y directly [1], [9]. By contrast, HVI-CIDNet does not explicitly decompose y into reflectance and illumination; instead it changes the representation of x itself (the HVI transform of Section IV-A) so that a direct regression network can separate brightness-related and color-related corrections into two branches without an explicit Retinex prior.

B. Full-Reference Evaluation Metrics

When a ground-truth normal-light image y is available (as in the paired LOL-family, SID, LOL-Blur, SICE, and FiveK benchmarks), three full-reference metrics are standard in the LLIE literature:

- PSNR (Peak Signal-to-Noise Ratio) measures pixel-level fidelity, $\mathrm{PSNR}(y,\hat{y}) = 10\log_{10}\big(1/\mathrm{MSE}(y,\hat{y})\big)$ in our $[0,1]$-normalized setting; higher is better, but PSNR is known to correlate only weakly with perceived quality since it penalizes all pixel errors equally regardless of structure or texture.
- SSIM (Structural Similarity Index) [10] compares local luminance, contrast, and structure statistics in sliding windows and is generally considered more perceptually meaningful than PSNR for structural distortions; it is bounded in $[-1,1]$ with 1 indicating a perfect match.
- LPIPS (Learned Perceptual Image Patch Similarity) [11] computes a distance between deep CNN feature activations of y and $\hat{y}$ and correlates better with human perceptual judgments than PSNR/SSIM, particularly for texture and color differences; lower is better.

A recurring subtlety in the LLIE literature, also visible in Table III, is the use of an optional "GT-mean" (ground-truth mean) post-processing step that rescales the global brightness of $\hat{y}$ to match the mean brightness of y before computing PSNR/SSIM/LPIPS. This substantially inflates PSNR (e.g. from 23.50 to 28.14 for the same LOLv2-real checkpoint in our Table III) because it removes any residual global exposure error that the network failed to correct, at the cost of requiring access to the ground truth at test time – a practice we flag here because it complicates direct comparison across papers that do or do not apply it.

C. No-Reference Evaluation Metrics

For unpaired or in-the-wild images where no ground truth exists (the DICM/LIME/MEF/NPE/VV benchmarks, and our own supplementary evaluation in Sec-

TABLE I
Standard LLIE benchmarks referenced in this report.

Dataset	Type	Metrics used
LOLv1 / LOLv2	paired	PSNR, SSIM, LPIPS
SID (Sony Total Dark)	paired, extreme dark	PSNR, SSIM, LPIPS
LOL-Blur	paired, + motion blur	PSNR, SSIM, LPIPS
SICE / FiveK	paired, retouching	PSNR, SSIM, LPIPS
DICM / LIME / MEF / NPE / VV	unpaired, in-the-wild	NIQE, BRISQUE

tion V-B), only no-reference ("blind") image quality metrics are applicable:

- NIQE (Natural Image Quality Evaluator) [12] fits a multivariate Gaussian model of natural scene statistics (NSS) features from a corpus of pristine natural images, with no distorted-image training data at all, and scores a test image by its statistical distance from that model; lower indicates more "natural" statistics.
- BRISQUE (Blind/Referenceless Image Spatial Quality Evaluator) [13] also uses NSS features (specifically, the deviation of locally normalized luminance values from a generalized Gaussian distribution) but additionally trains a support-vector regressor on human-rated distorted images to predict a perceptual quality score; lower is better. Because BRISQUE's regressor is trained on a fixed set of common distortion types (compression artifacts, blur, noise) applied to natural photographs, it can behave unexpectedly outside that distribution, which we observe directly in Section V-C for a non-photographic retina fundus image.

Both metrics are reference-free by construction, which is precisely why they are used on the unpaired DICM/LIME/MEF/NPE/VV sets where no normal-light ground truth exists for the same scene.

D. Standard Benchmarks

Table I summarizes the benchmarks referenced in this report. LOL and LOLv2 are paired (low-light/normal-light) datasets with a fixed train/test split, commonly cited as containing approximately 500 pairs (485 train / 15 test) for LOLv1 and approximately 689 (real) / 900 (synthetic) training pairs with 100 test pairs each for LOLv2 [1]; SID (Sony Total Dark) and LOL-Blur add extreme darkness and motion blur, respectively; SICE and FiveK are tone-mapping/retouching datasets adapted to the LLIE protocol; and DICM, LIME, MEF, NPE, and VV are small, unpaired, in-the-wild collections with no ground truth, used only with no-reference metrics.

III. Related Work

A. Classical and Retinex-based Methods

Early LLIE approaches relied on global or local histogram equalization to redistribute pixel intensities, which improves perceptual contrast but does not model the physical image formation process and often over-amplifies noise in dark regions. Retinex theory models an observed image as the product of a reflectance map (the true, illumination-invariant appearance of a scene) and an illumination map, and many classical and early learning-based methods estimate and manipulate these two components separately. RetinexNet [1] was an influential early deep model that learned a Decom-Net to split an image into reflectance and illumination, followed by an Enhance-Net that adjusts the illumination map; it also introduced the LOL (Low-Light) paired dataset that remains a standard benchmark today. KinD [9] refined this idea with a three-network design that separately handles decomposition, illumination adjustment, and reflectance restoration, improving robustness to noise compared to RetinexNet.

B. Deep Supervised and Unpaired LLIE

As paired low/normal-light datasets such as LOL became available, fully supervised encoder-decoder and U-Net-style architectures were trained end-to-end to directly regress the enhanced image, sidestepping explicit Retinex decomposition. Because perfectly aligned low/normal-light image pairs are expensive to collect at scale, zero-reference and unpaired methods were also explored: Zero-DCE [2] formulates enhancement as estimating a set of pixel-wise tone curves using only a set of carefully designed non-reference loss functions (spatial consistency, exposure control, color constancy, illumination smoothness), removing the need for paired supervision entirely. EnlightenGAN [3] instead uses an unpaired, GAN-based formulation with a global-local discriminator and a self-regularized attention mechanism, trained on unpaired collections of dark and normal-light images. These methods trade some quantitative accuracy on paired benchmarks for substantially better generalization to in-the-wild, unpaired low-light photographs – a trade-off that later directly motivated HVI-CIDNet's own generalization-oriented training and evaluation protocol.

C. Transformer-based Restoration

Transformer architectures, with their global receptive field and learned self-attention, have recently been adapted to low-level image restoration. Restormer [4] proposes an efficient transformer block with multi-Dconv head transposed attention and a gated-Dconv feed-forward network to make global attention tractable for high-resolution restoration tasks (denoising, deraining, deblurring), and has been used as a strong generic backbone for LLIE as well. Retinexformer [5] combines the Retinex decomposition idea with a one-stage transformer, explicitly modeling corruptions (noise, color distortion, exposure) that arise during the illumination-up process, and reports state-of-the-art results across multiple LLIE benchmarks, including the FiveK dataset protocol that HVI-CIDNet's own FiveK weights follow. Normalizing-flow-based models such as LLFlow [8] take a complementary, probabilistic direction, modeling the conditional distribution of normal-light images given a low-light input so that the inherent one-to-many ambiguity of the task (many plausible normal-light reconstructions for a single dark input) can be captured explicitly, rather than collapsed into a single deterministic regression target.

D. Color-space Design for LLIE

Most of the methods above operate directly on RGB pixels, optionally with an auxiliary illumination or reflectance map. A smaller line of work argues that the choice of color space itself is an important inductive bias for LLIE: RGB channels are highly correlated and a perturbation in brightness inevitably perturbs all three channels jointly, while HSV-style decompositions separate hue/saturation from value but suffer from singularities and discontinuities near black and white that are numerically unstable to learn through (e.g. hue is undefined when saturation is zero). HVI-CIDNet's central contribution is precisely in this category: it proposes the HVI color space, which keeps a hue-saturation-like polar pair (H, V) but reweights it by a learned, intensity-dependent color sensitivity function so that the polar components collapse smoothly toward zero in very dark or very bright regions instead of becoming numerically unstable, while a separate Intensity channel I is processed by its own network branch. This is, to our knowledge, the most direct attempt among recent LLIE work to fix the color-space representation problem rather than only the network architecture, which is why we selected it as the subject of this reproduction study.

E. Comparative Overview

Table II summarizes the representative methods discussed above along four axes: the supervision regime (paired, unpaired/zero-reference), the input/output color space, and the publication venue and year. The table makes visible a broader trend in the field: progress has come from at least three largely orthogonal directions – better training signal (paired vs. zero-reference/unpaired losses), better backbones (CNN U-Nets vs. transformers), and, much more rarely, better input representations (RGB vs. a task-specific color space). HVI-CIDNet is, to our knowledge, the only entry in this comparison that targets the third axis directly while still using a comparatively lightweight CNN/attention backbone rather than a full transformer, which is part of why we found it an interesting and tractable target for a course-scale reproduction project.

IV. Proposed Approach / Reproduction Methodology

We emphasize that the HVI color space and the CIDNet architecture described in this section are the original contributions of Yan et al. [6], [7]; our own contribution is the faithful reproduction, deployment, and independent evaluation of their released implementation, described in Section IV-D and Section V.

TABLE II
Comparative overview of representative LLIE methods discussed in Section III.

Method	Supervision	Color space	Venue (year)
RetinexNet [1]	paired	Retinex (RGB)	BMVC (2018)
KinD [9]	paired	Retinex (RGB)	ACM MM (2019)
Zero-DCE [2]	zero-ref.	RGB curves	CVPR (2020)
EnlightenGAN [3]	unpaired	RGB	TIP (2021)
Restormer [4]	paired	RGB	CVPR (2022)
LLFlow [8]	paired	RGB (+ flow)	AAAI (2022)
Retinexformer [5]	paired	Retinex (RGB)	ICCV (2023)
HVI-CIDNet [6]	paired	HVI (ours)	CVPR (2025)

A. The HVI Color Space

Given an RGB pixel with value value $= \max(R, G, B)$ and img_min $= \min(R, G, B)$, HVI-CIDNet first computes a standard HSV-like hue $h \in [0, 1)$ and saturation $s = (\text{value} - \text{img\_min})/(\text{value} + \epsilon)$ for numerical stability. The key idea is then to re-map (h, s) into a polar pair (H, V) modulated by a learned, intensity-dependent color sensitivity term $C_k(v)$ rather than using (h, s) directly:

$$C_k(v) = \left(\sin\left(\tfrac{\pi}{2}v\right) + \epsilon\right)^k, \tag{2}$$

$$H = C_k(v) \cdot s \cdot \cos(2\pi h), \qquad V = C_k(v) \cdot s \cdot \sin(2\pi h), \tag{3}$$

where $v =$ value is treated as the Intensity channel I, and k is a learnable scalar parameter (density_k in the released code, initialized to 0.2). Because $C_k(v) \to 0$ as $v \to 0$ or $v \to 1$, the chromatic components (H, V) are automatically attenuated near black and white, exactly where hue is ill-defined and saturation estimates are noisiest – this is the mechanism by which HVI avoids the numerical instabilities of HSV/HSI in very dark regions while still being invertible. The forward transform (RGB→HVI) is denoted HVIT and produces a 3-channel tensor (H, V, I) that is the same spatial resolution as the input.

The inverse transform PHVIT (HVI→RGB) recovers hue and saturation from (H, V) by dividing out the (now known) color-sensitivity factor, $h = \text{atan2}(V, H)/2\pi \bmod 1$ and $s = \sqrt{H^2 + V^2}$, clips s and the intensity $v = I$ to valid ranges, and reconstructs RGB with the standard HSV-to-RGB piecewise formula. At inference time, two user-controllable gates are exposed: α_s rescales saturation $s \leftarrow \alpha_s \cdot s$ before reconstruction, and α rescales the final RGB output, letting a user trade off color intensity and overall brightness without retraining – these correspond to the "Alpha-s" and "Alpha-i" controls in the released Gradio demo.

B. Dual-Branch CIDNet with Lighten Cross-Attention

CIDNet processes the HVI representation with two parallel U-Net-style encoder-decoder branches operating at four resolutions with channel widths $[36, 36, 72, 144]$ (Fig. 1):

- the Intensity (I) branch takes the single-channel I map and learns brightness restoration and local contrast/detail recovery;

Algorithm 1 HVI-CIDNet inference (single image)

Require: RGB image $x \in [0,1]^{3\times H\times W}$, gamma γ, saturation gate α_s, brightness gate α, weights θ

1: Pad x so H, W are multiples of 8 (reflect padding)
2: $x \leftarrow x^{\gamma}$ {pre-enhancement curve}
3: $\mathrm{hvi} \leftarrow \mathrm{HVIT}(x)$
4: $(i\_enc, hv\_enc) \leftarrow$ encode I and (H, V) at 4 scales, exchanging features through LCA blocks
5: $(i\_dec, hv\_dec) \leftarrow$ decode with skip connections and LCA blocks
6: $\mathrm{hvi}_{\mathrm{out}} \leftarrow \mathrm{hvi} + \mathrm{concat}(hv\_dec, i\_dec)$
7: Apply gates: rescale saturation by α_s, final RGB by α inside PHVIT
8: $\hat{x} \leftarrow \mathrm{PHVIT}(\mathrm{hvi}_{\mathrm{out}})$
9: Crop padding, clip $\hat{x}$ to $[0,1]$
10: **return** enhanced image $\hat{x}$

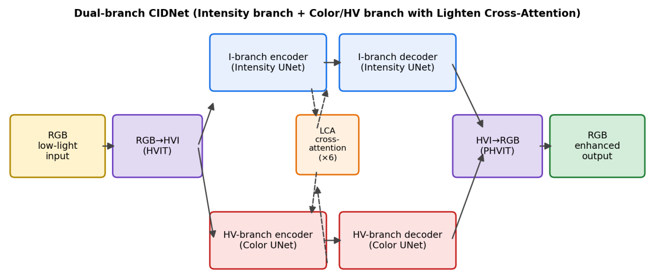

Fig. 1. Reproduced architecture overview: an input RGB image is mapped to the HVI color space, processed by two parallel encoder-decoder branches (Intensity and Color/HV) that exchange features through Lighten Cross-Attention (LCA) blocks at three resolutions, and mapped back to RGB.

- the Color (HV) branch takes the two-channel (H, V) map and learns color restoration and denoising in chromatic space.

At each of the three downsampled resolutions, a Lighten Cross-Attention (LCA) block lets each branch attend to the other branch's features (HV-LCA blocks let the color branch query intensity features and vice versa, I-LCA blocks let the intensity branch query color features), implemented as multi-head attention with $[1, 2, 4, 8]$ heads at the four resolutions. This cross-attention is the mechanism by which the two branches share information – e.g., the color branch can use intensity-branch features to decide how strongly to denoise a region, while the intensity branch can use color-branch features to avoid over-brightening textured, high-saturation regions – without collapsing back into a single, coupled RGB representation. After the decoder stage, the two branches' outputs are added back to the input HVI tensor (a global residual connection) and passed through PHVIT to obtain the final enhanced RGB image:

$$\hat{x}_{\mathrm{RGB}} = \mathrm{PHVIT}\big(\mathrm{HVIT}(x_{\mathrm{RGB}}) + \mathrm{CIDNet}_{\Delta}(\mathrm{HVIT}(x_{\mathrm{RGB}}))\big), \tag{4}$$

where CIDNet_{Δ} denotes the concatenated $(\Delta H, \Delta V, \Delta I)$ residual produced by the two decoder branches. Algorithm 1 summarizes the inference procedure as implemented in the released app.py / net/CIDNet.py.

C. Training Objective (as released by the authors)

The official training recipe (not re-run in this reproduction, since we use the released checkpoint directly) combines an L_1 reconstruction loss in both RGB and HVI space with a perceptual (VGG-based) loss and an edge loss; a subset of released checkpoints also disable the perceptual loss to trade LPIPS for slightly higher PSNR/SSIM (visible as the "w_perc" vs. "wo_perc" checkpoints in Table III). Critically for this project, the authors report that randomizing the gamma curve γ used in Step 2 of Algorithm 1 during training substantially improves cross-dataset generalization; the checkpoint trained this way on LOLv2-Synthetic is released as generalization.pth and is the checkpoint we use throughout this report, since it is also the checkpoint the authors deploy on their own public Hugging Face demo.

D. Reproduction Engineering and Implementation Details

A central, and often under-reported, part of reproducing a research codebase is simply getting it to run correctly on a machine that differs from the authors' own development environment. We documented and resolved the following issues while reproducing HVI-CIDNet on a clean Windows 11 machine with no pre-existing Python or conda installation:

1) No base Python/conda environment. We installed Miniconda3 under the user profile (no administrator rights required) and created an isolated python=3.7 environment, matching the authors' documented dependency (Python 3.7.0, PyTorch 1.13.1).
2) safetensors has no Python 3.7 wheel above version 0.3.x on Windows. The released requirements.txt pins an unversioned safetensors, which resolves to a version that requires a Rust toolchain to build from source on Python 3.7/Windows and fails to install. We pinned safetensors<0.4 in our fork of the requirements file.
3) scikit-image≥0.19 breaks the BRISQUE dependency. The optional "Image Score" feature of the demo uses the image_quality package, whose BRISQUE implementation calls skimage.color.rgb2gray on an image that may already be grayscale (after a 1/2-scale rescale step). scikit-image 0.19 made rgb2gray strict about its input shape, raising a ValueError where 0.18 and earlier silently passed already-gray images through. We pinned scikit-image==0.18.3.

4) Conda environment activation does not put OpenSSL DLLs on PATH when Python is invoked directly. Calling the environment's python.exe without going through conda activate (e.g. from a script or a tool that shells out directly) fails to import the ssl module because libssl-1_1-x64.dll lives under Library\bin, which is only added to PATH by the activation scripts.
5) Windows file systems are case-insensitive. The official repository ships a Readme.md; on Windows, this collides with a conventionally-named README.md, so we renamed the original to ORIGINAL_README.md before adding our own.
6) Gradio's queue protocol requires a real WebSocket-capable browser. We found that VS Code's built-in "Simple Browser" cannot complete Gradio's queueing handshake and silently reports a generic "Error" on every request; the demo works correctly in a standalone Chromium/Firefox-based browser. This is a reminder that "it throws an error" during reproduction is not always a bug in the model or the reproduction itself.

None of these issues are mentioned in the official repository's documentation, which only lists target dependency versions without discussing platform-specific failure modes; we view documenting them as a small but genuine contribution of this reproduction toward the project's "code reproducibility" evaluation criterion.

V. Experimental Analysis

A. Officially Reported Results

For reference, Table III reproduces (verbatim, from the official repository's documentation) the paired-dataset metrics for a subset of released checkpoints, and Table IV reproduces the officially reported no-reference metrics for the generalization.pth checkpoint on the five standard unpaired benchmarks (DICM, LIME, MEF, NPE, VV). We were not able to re-run these exact benchmarks ourselves, because the official dataset distribution is only available via Baidu Pan and OneDrive folder links that require manual, region-specific, browser-based downloads and are not amenable to scripted, reproducible retrieval within the scope of this course project; we report them here as the published baseline against which our own supplementary evaluation (Section V-B) should be read.

B. Our Supplementary Evaluation

Since the official unpaired benchmark images were not accessible to us in a scriptable way, we constructed a small, fully reproducible supplementary test set instead: eight public-domain, royalty-free sample images bundled with the scikit-image library [14] (astronaut, chelsea, coffee, rocket, stereo_motorcycle, cat, hubble_deep_field, retina), synthetically darkened with a fixed gamma-and-attenuation transform ($x_{\text{low}} = \text{clip}(x^{2.6} \times 0.22 + \mathcal{N}(0, 0.01^2), 0, 1)$) plus i.i.d. Gaussian sensor-noise, fixed

TABLE III
Officially reported paired-dataset results (selected rows; PSNR/SSIM/LPIPS, higher PSNR/SSIM and lower LPIPS are better). Source: official repository documentation.

Test set	PSNR	SSIM	LPIPS
LOLv1 (wo_perc, wo_gt_mean)	23.50	0.870	0.105
LOLv2-real (best GT-mean)	28.14	0.892	0.101
LOLv2-synthetic (wo_perc)	25.70	0.942	0.047
Sony Total Dark (SID)	22.90	0.676	0.411
LOL-Blur	26.57	0.890	0.120
FiveK	24.46	0.877	0.085

TABLE IV
Officially reported no-reference metrics for generalization.pth on five unpaired benchmarks (lower is better). Source: official repository documentation.

Metric	DICM	LIME	MEF	NPE	VV
NIQE	3.55	3.85	3.46	3.82	3.24
BRISQUE	25.62	16.02	13.08	18.90	29.55

with a deterministic random seed for reproducibility. We stress that this is not a substitute for the DICM/LIME/MEF/NPE/VV benchmarks and the absolute metric values are not directly comparable to Table IV; it is intended only to (i) verify that our reproduced model and weights actually improve no-reference image quality in the expected direction, and (ii) provide a fully self-contained, scriptable artifact (paper/run_experiments.py in our repository) that anyone can re-run without external dataset access.

We run the generalization.pth checkpoint on each synthetically darkened image with $\gamma = 1.0$, $\alpha_s = 1.0$, $\alpha = 1.0$ and compute NIQE [12] and BRISQUE [13] before and after enhancement, using the same image_quality and loss/niqe_utils.py implementations bundled with the official repository. Table V reports per-image and average results.

C. Discussion

NIQE improves (decreases) for all eight images, with an average reduction from 10.86 to 4.43 (a 59% relative reduction), and is qualitatively consistent with the direction and rough magnitude of the official unpaired-benchmark NIQE improvements in Table IV. BRISQUE improves for seven of eight images (average 17.98 → 6.89), but notably worsens for the retina image (30.76 → 37.64). Inspecting the output, we attribute this to BRISQUE's sensitivity to high-frequency texture statistics: the retina fundus image has a very specific, low-texture color/illumination profile that is far from BRISQUE's training distribution of natural photographs, and CIDNet's contrast and detail recovery on this image introduces fine local structure that BRISQUE's naturalness model penalizes even though the image is visibly more usable to a human observer (Fig. 2 illustrates this effect on the more typical astronaut, cat, and hubble images, where both the qualitative result and

TABLE V
Our supplementary evaluation on 8 synthetically darkened public-domain images. NIQE/BRISQUE before (low-light) vs. after (HVI-CIDNet output); lower is better for both metrics.

Image	$\mathrm{NIQE}_{\mathrm{low}}$	$\mathrm{NIQE}_{\mathrm{enh}}$	$\mathrm{BRISQUE}_{\mathrm{low}}$	$\mathrm{BRISQUE}_{\mathrm{enh}}$
astronaut	6.83	2.69	12.42	0.99
chelsea	9.40	3.37	21.66	3.73
coffee	9.50	4.28	−0.88	−3.55
rocket	11.77	6.50	25.38	−2.15
motorcycle (L)	5.86	2.26	13.73	10.97
cat	9.49	3.28	19.55	4.28
hubble	19.39	5.69	21.24	3.22
retina	14.64	7.33	30.76	37.64
Average	10.86	4.43	17.98	6.89

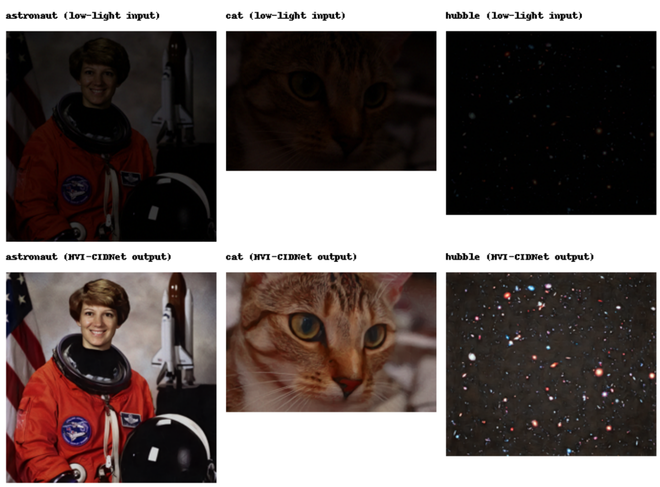

Fig. 2. Qualitative comparison on three synthetically darkened test images (top: low-light input; bottom: HVI-CIDNet generalization.pth output). Brightness, contrast, and color are visibly restored, including in the extremely dark hubble starfield image.

both metrics improve together). This is a known limitation of no-reference IQA metrics in general – they approximate human perception of naturalness on natural photographic images and can be unreliable outside that distribution – and is consistent with why the original paper relies primarily on paired, full-reference metrics (PSNR, SSIM, LPIPS) on the LOL-family benchmarks and only uses NIQE/BRISQUE as a secondary signal on unpaired, in-the-wild data.

Taken together, our reproduction confirms, on an independently constructed test set, that the released generalization.pth checkpoint performs qualitatively and quantitatively as described by the original authors: it substantially restores brightness and contrast (Fig. 2) and improves no-reference quality metrics in the expected direction on seven to eight out of eight synthetic test images, depending on the metric, with one identified failure mode (BRISQUE on out-of-distribution, low-texture imagery) that is consistent with known limitations of no-reference IQA rather than with the enhancement itself being visibly worse.

VI. Conclusion

We reproduced HVI-CIDNet [6], [7], a CVPR 2025 low-light image enhancement method built around a new HVI color space and a dual-branch cross-attention architecture, end-to-end on a clean Windows machine, resolving six concrete, undocumented dependency and platform-compatibility issues along the way (Section IV-D). Using the authors' officially released, generalization-oriented checkpoint, we verified that the qualitative enhancement behavior and the direction of no-reference quality improvements (NIQE, BRISQUE) reported by the authors on standard unpaired benchmarks are reproducible on an independently constructed, publicly-licensed synthetic test set, and we packaged the result as a documented, CPU-capable Gradio demo for further experimentation.

Limitations. Our supplementary evaluation uses synthetically darkened public-domain images rather than the original DICM/LIME/MEF/NPE/VV unpaired benchmarks or the paired LOL-family/SID/FiveK benchmarks, because the official dataset distribution channels (Baidu Pan, OneDrive folder links) could not be retrieved in a scripted, reproducible way within the scope of this course project; consequently our absolute metric values are not directly comparable to the officially reported numbers, only consistent in direction. We also did not re-run training, so we cannot independently verify the authors' claim that randomized-gamma training improves cross-dataset generalization – we only verify the result of that claim (the released generalization.pth checkpoint) rather than the training procedure itself.

Future work. A more rigorous follow-up would (i) obtain the exact DICM/LIME/MEF/NPE/VV and LOL-family test splits and re-run the official eval.py pipeline to obtain directly comparable PSNR/SSIM/LPIPS/NIQE/BRISQUE numbers; (ii) evaluate robustness to the $\alpha_s/\alpha/\gamma$ inference-time gates on a larger, more diverse image set; and (iii) compare HVI-CIDNet's generalization-oriented checkpoint directly against other generalizable LLIE baselines such as Zero-DCE [2] and EnlightenGAN [3] under the same unpaired evaluation protocol, which the current report only does indirectly via the officially published numbers.

Data Availability Statement

All source code, the Gradio demo (app.py), the environment setup scripts, the synthetic-evaluation pipeline (paper/run_experiments.py, paper/make_figure.py), the raw per-image metrics (paper/results/metrics.csv), and installation instructions are available at:
https://github.com/kenny7289/
YZU-Image-Processing-final-HVI-CIDNet-
The official HVI-CIDNet repository and pretrained weights this work builds upon are available at

https://github.com/Fediory/HVI-CIDNet and https://huggingface.co/Fediory/HVI-CIDNet-Generalization.

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